



YOLO-World: Real-Time Open-Vocabulary Object Detection

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Introduction

전통적인 컴퓨터 비전 인식(perception) task

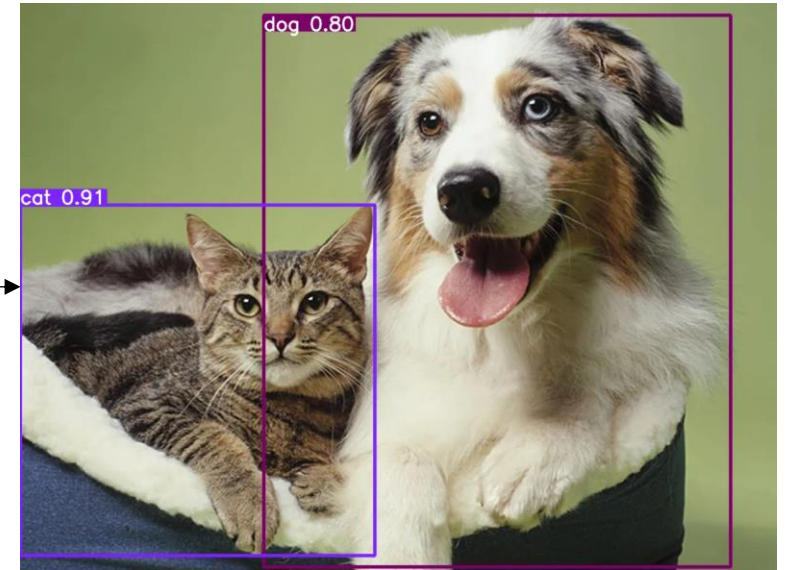
- 미리 정의된 특정 카테고리만 인식 가능
- 새 클래스를 추가하려면 모델 재훈련 필요



Input image

Conventional
Object detector

YOLOv8

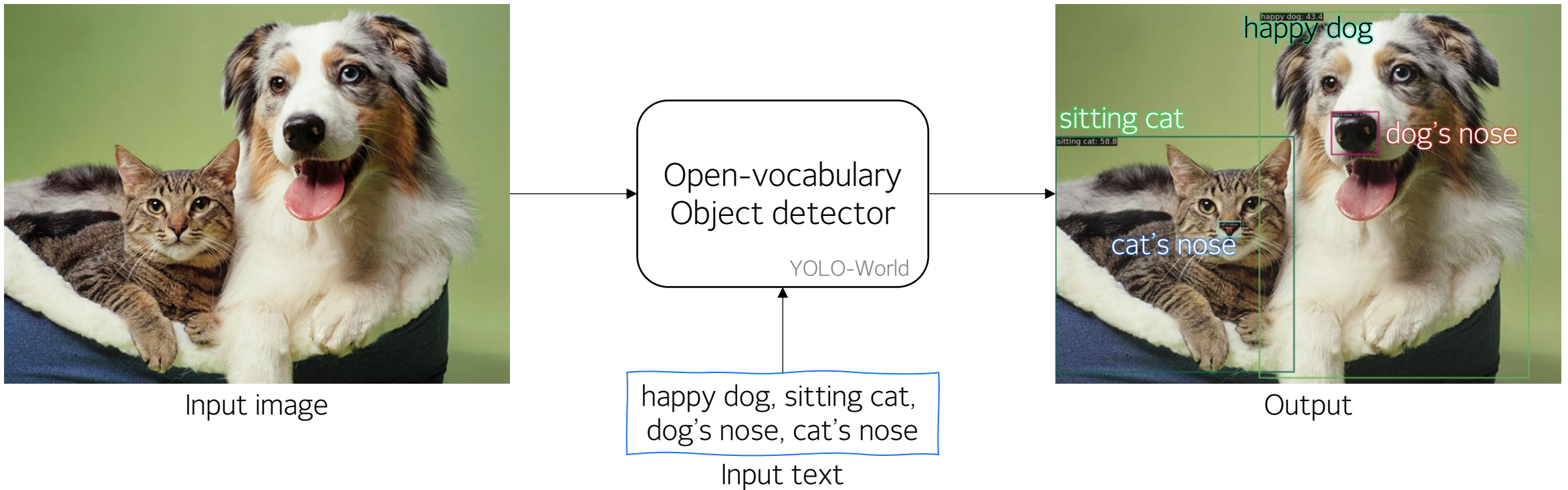


Output

Introduction

Open-vocabulary task

- 사전에 학습된 카테고리에 제한되지 않고 임의의 텍스트 입력을 통해 다양한 객체 인식 가능
- 새 클래스에 대해 즉시 인식 가능



Introduction

YOLO-World

- 기존 YOLO series의 한계(전통적인 Object Detector)에서 벗어나고자 함
→ Open-vocabulary Object Detector(OVD)
- 이전의 OVD는 모델이 무거워 실제 환경(e.g. 엣지디바이스)에서 사용이 어려움
→ YOLO-World는 실시간 추론 및 하위 애플리케이션 배포 용이

Related works:

Traditional Object detection

- 미리 정의된 고정된 어휘(Fixed vocabulary)에 속하는 객체만 검출
- Region-based method: e.g. Faster R-CNN
- Pixel-based method: e.g. SSD, YOLO
- Query-based method: e.g. DETR
- YOLO series

Related works

Open-vocabulary Object Detection

- Knowledge distillation 방식:
 - Base 클래스만 학습 후 Novel 클래스 평가 (OV-COCO)
 - 언어 모델(BERT 등)로부터 지식 증류(distillation) 방식
- Region-text contrastive pre-training 계열:
 - 이미지-텍스트 매칭(region-text pairs)을 대규모로 학습
 - GLIP, Grounding DINO, RegionCLIP 등
- 한계: 무거운 탐지 모델 or 백본을 사용하여 높은 계산량 및 실제 환경 배포 제한

Method

Pre-training Formulation: Region-Text Pair

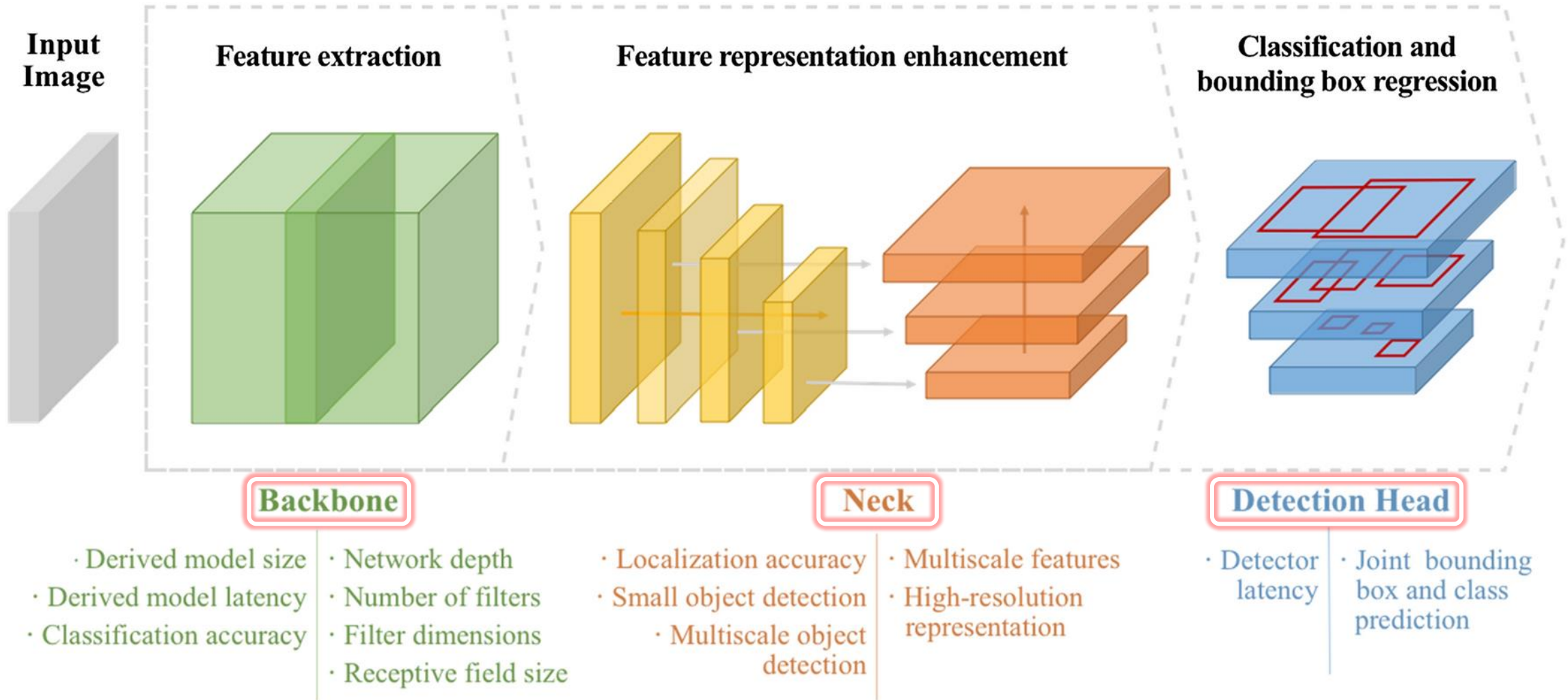
- YOLO-series annotation: $\{B_i, c_i\}_{i=1}^N$
 - i : instance
 - B_i : Bounding box
 - c_i : category label
- YOLO-World annotation: $\{B_i, t_i\}_{i=1}^N$
 - t_i : text for the region B_i
 - category name, noun phrases, or object descriptions
 - e.g. 'apple', 'a red apple', 'a red apple on the table'

Method

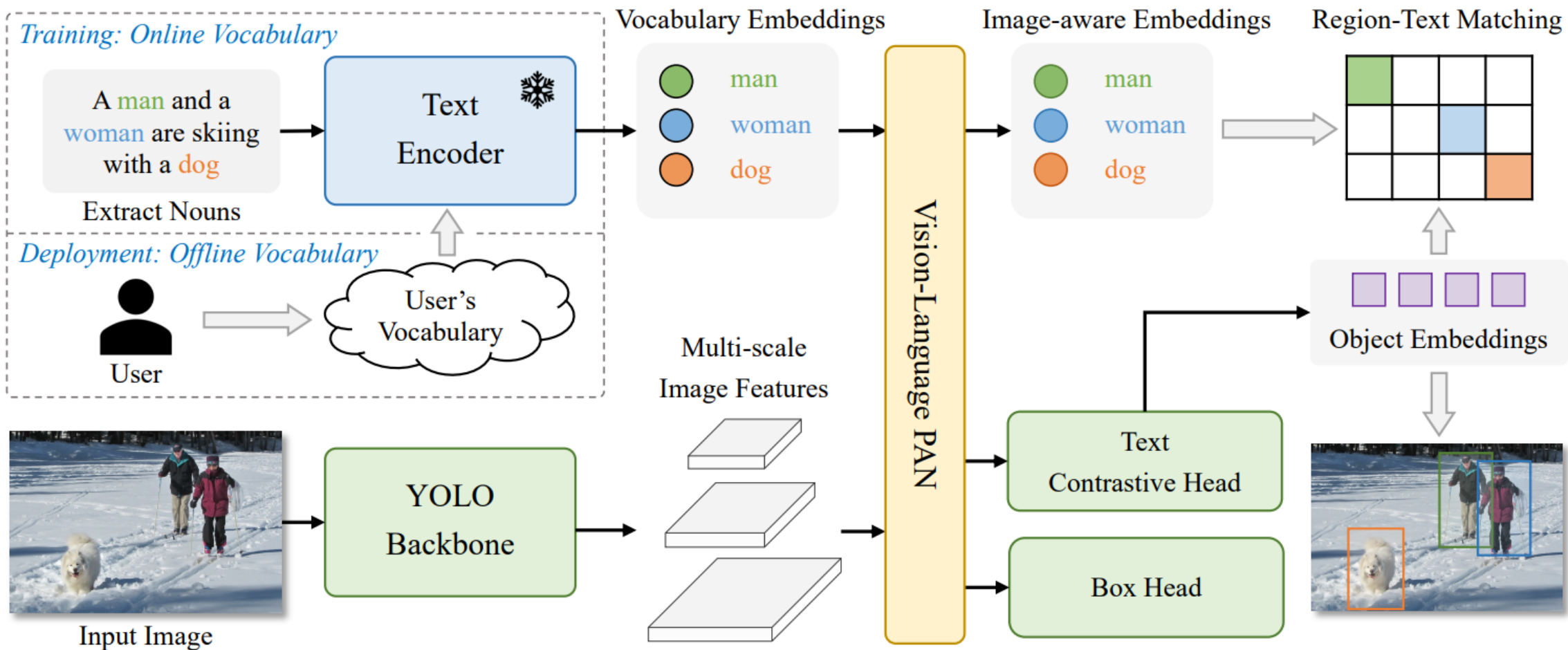
Pre-training Formulation: Region-Text Pair

- Input:
 - image I
 - texts T (a set of nouns)
- Output:
 - predicted boxes $\{\hat{B}_k\}$
 - The corresponding object embeddings $\{e_k\}$ $e_k \in \mathbb{R}^D$

Preliminaries: object detection 기본 구조

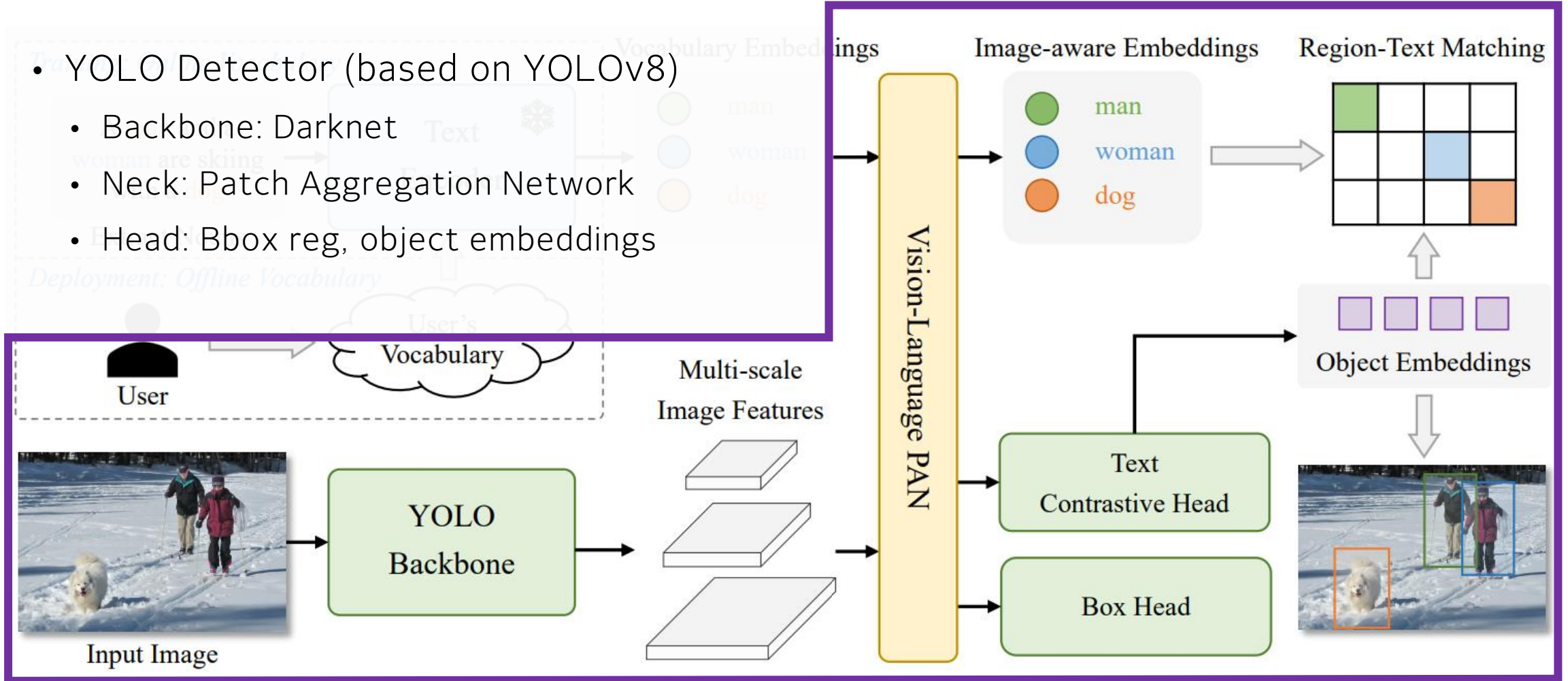


Method

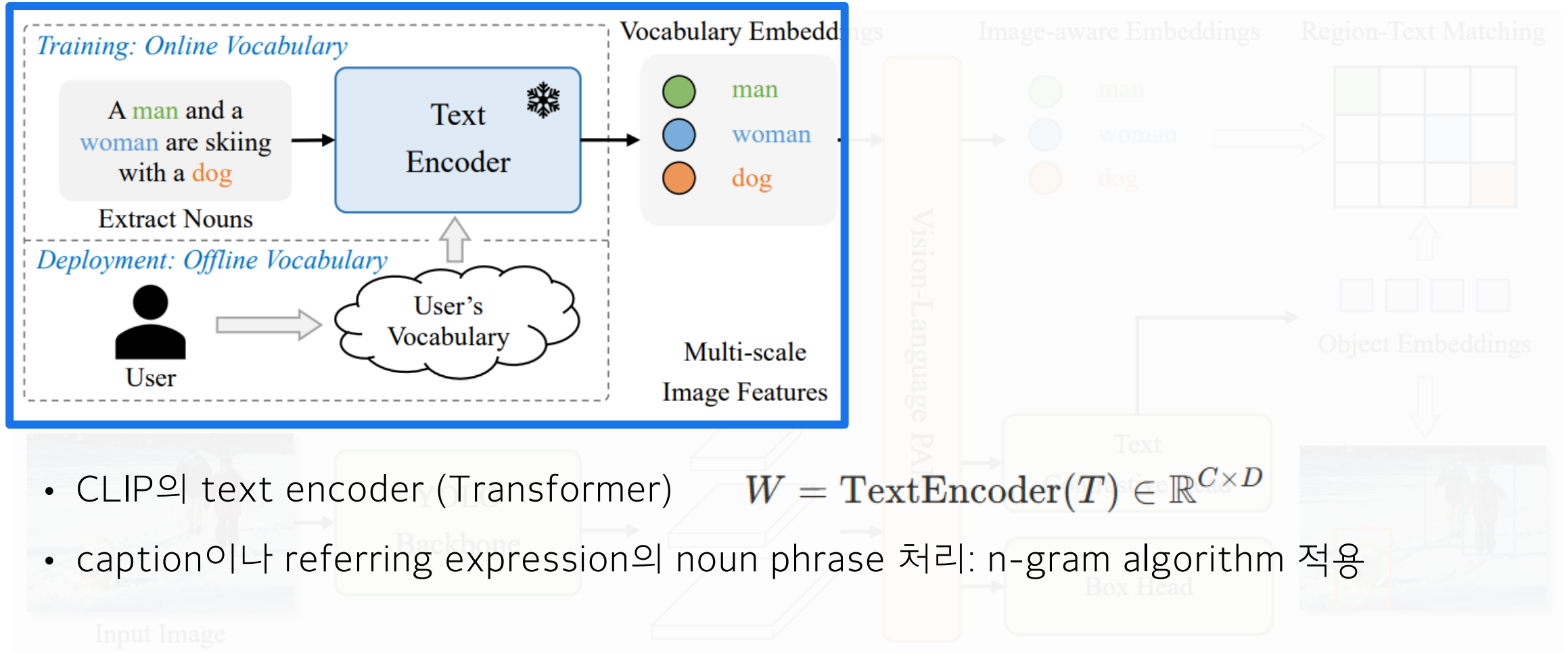


Method

- YOLO Detector (based on YOLOv8)
 - Backbone: Darknet
 - Neck: Patch Aggregation Network
 - Head: Bbox reg, object embeddings



Method



Method

- Text Contrastive Head

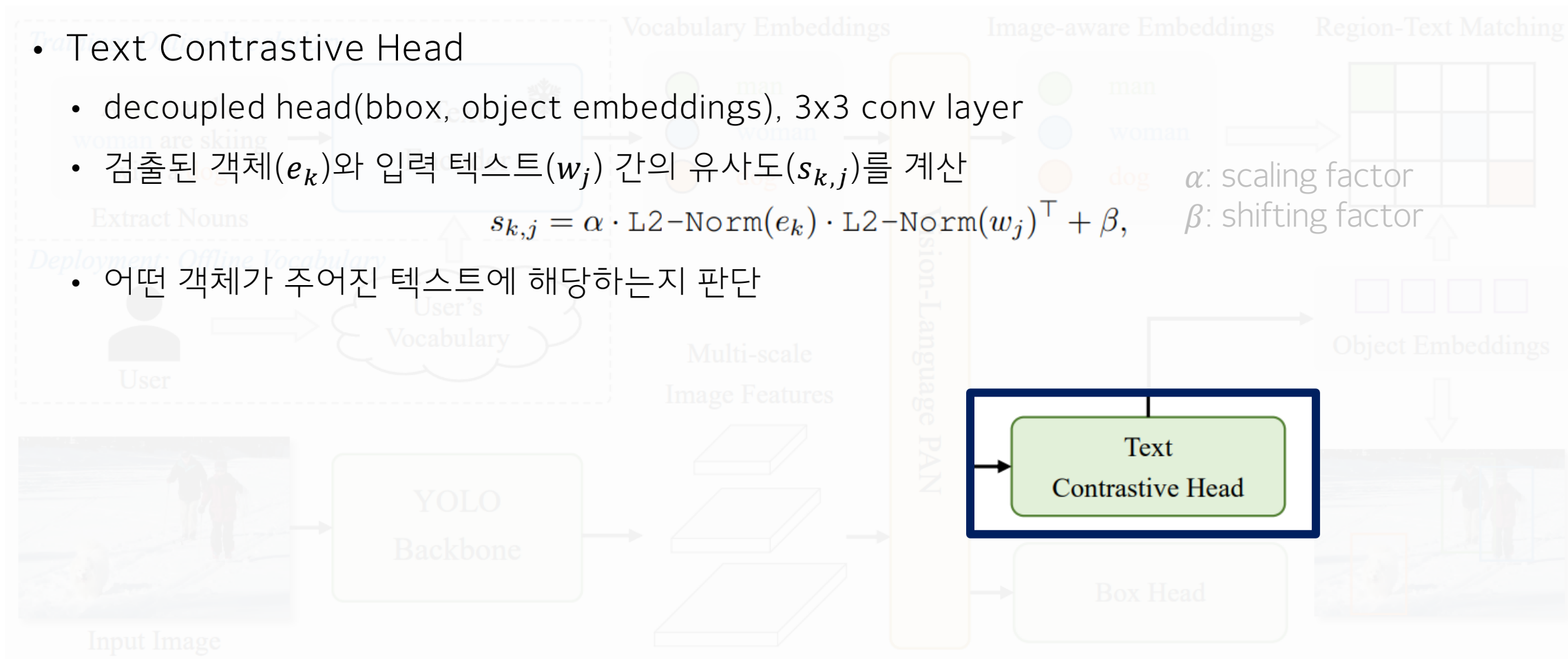
- decoupled head(bbox, object embeddings), 3x3 conv layer

- 검출된 객체(e_k)와 입력 텍스트(w_j) 간의 유사도($s_{k,j}$)를 계산

$$s_{k,j} = \alpha \cdot \text{L2-Norm}(e_k) \cdot \text{L2-Norm}(w_j)^T + \beta,$$

α : scaling factor
 β : shifting factor

- 어떤 객체가 주어진 텍스트에 해당하는지 판단



Method

- Training with Online Vocabulary

- Iteration 마다 mosaic sample(4 images)에 대해 동적으로 vocabulary 구성 (Online vocabulary T)
- Mosaic sample에 포함된 이미지의 모든 positive nouns를 포함
- 일부 random negative nouns 포함
- 최대 80개의 vocabulary size



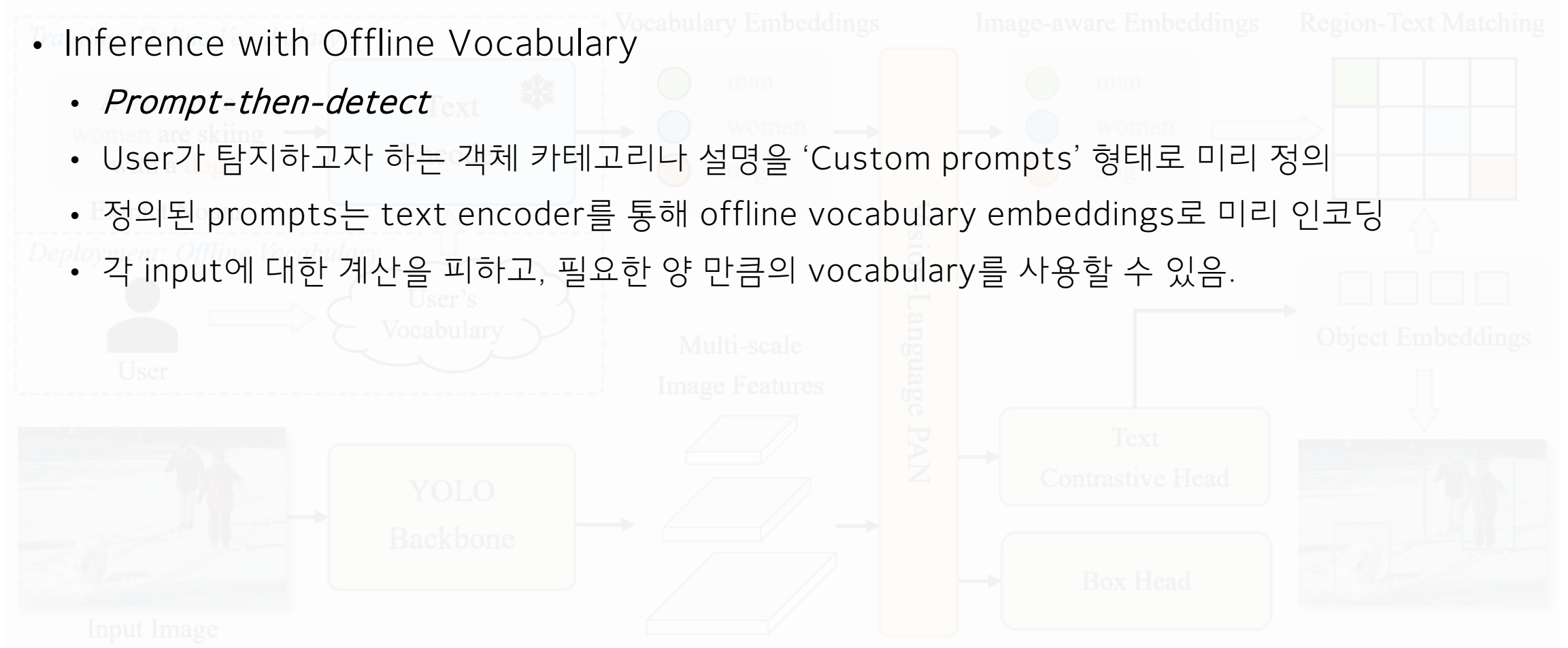
a) Sample images for Mosaic augmentation

Method

- Inference with Offline Vocabulary

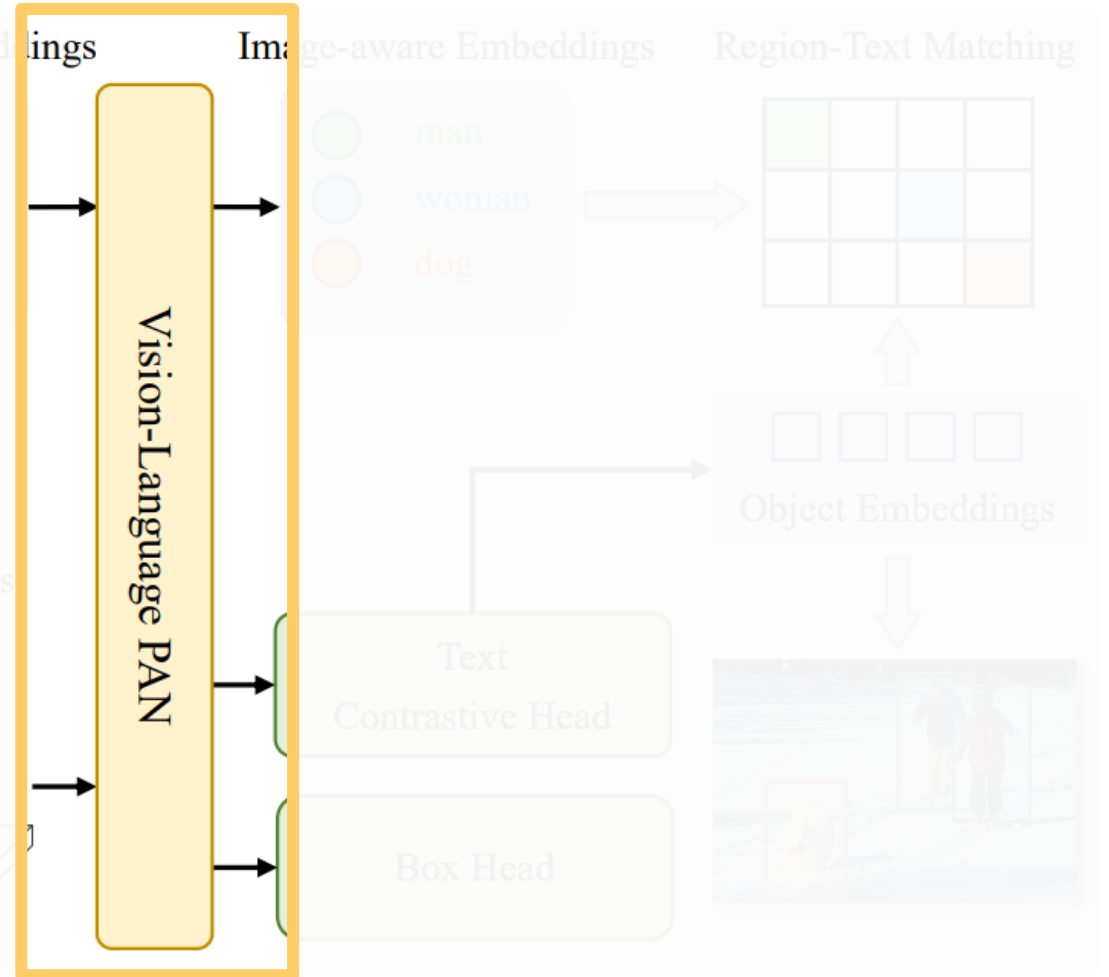
- *Prompt-then-detect*

- User가 탐지하고자 하는 객체 카테고리나 설명을 'Custom prompts' 형태로 미리 정의
 - 정의된 prompts는 text encoder를 통해 offline vocabulary embeddings로 미리 인코딩
 - 각 input에 대한 계산을 피하고, 필요한 양 만큼의 vocabulary를 사용할 수 있음.



Method

- Re-parameterizable Vision-Language PAN
 - YOLO 모델 내 image feature와 text embedding을 효과적으로 결합
 - Open-vocabulary object detection 성능 향상

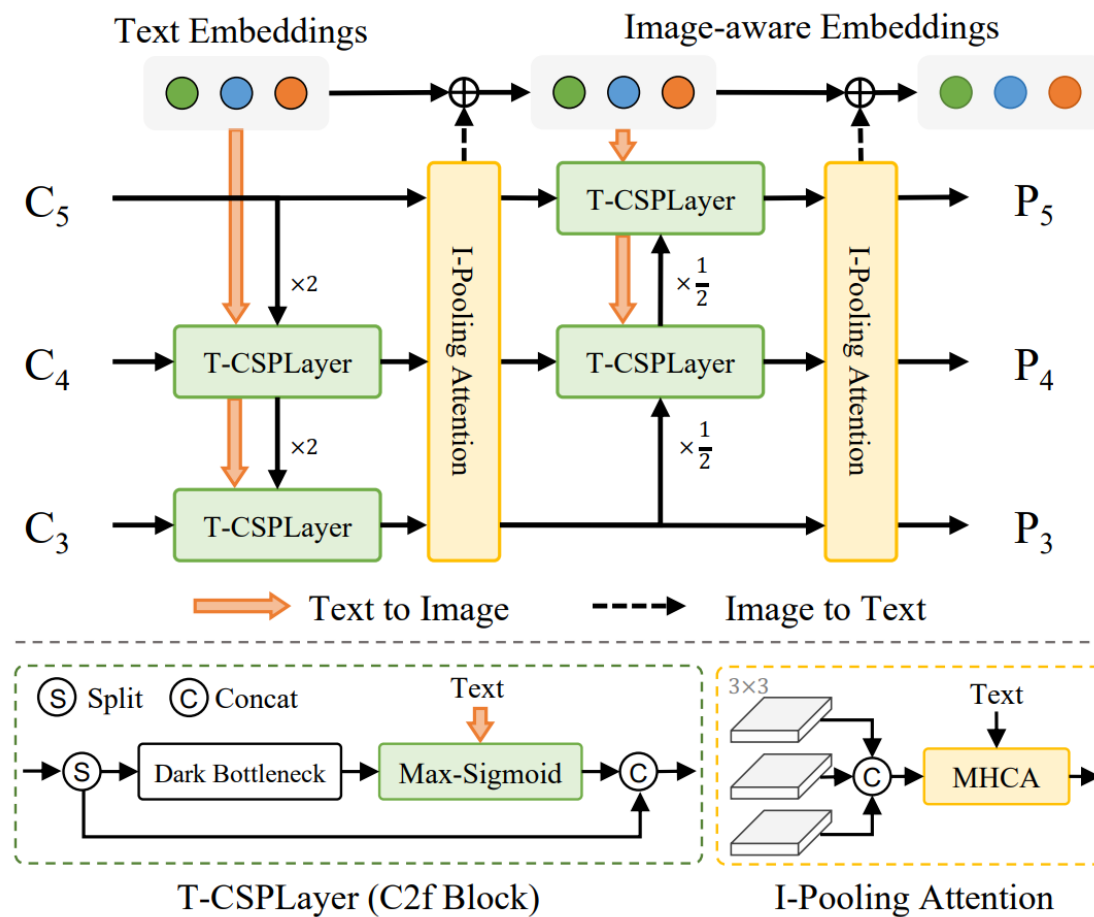


Method

- C_3, C_4, C_5 : multi-scale image features
- P_3, P_4, P_5 : feature pyramids

- Text-guided CSP layer
 - Multi-scale feature에 언어 정보 주입
- Image Pooling Attention
 - Text embedding을 이미지 정보로 강화

Interaction between Image feature & text feature



Method

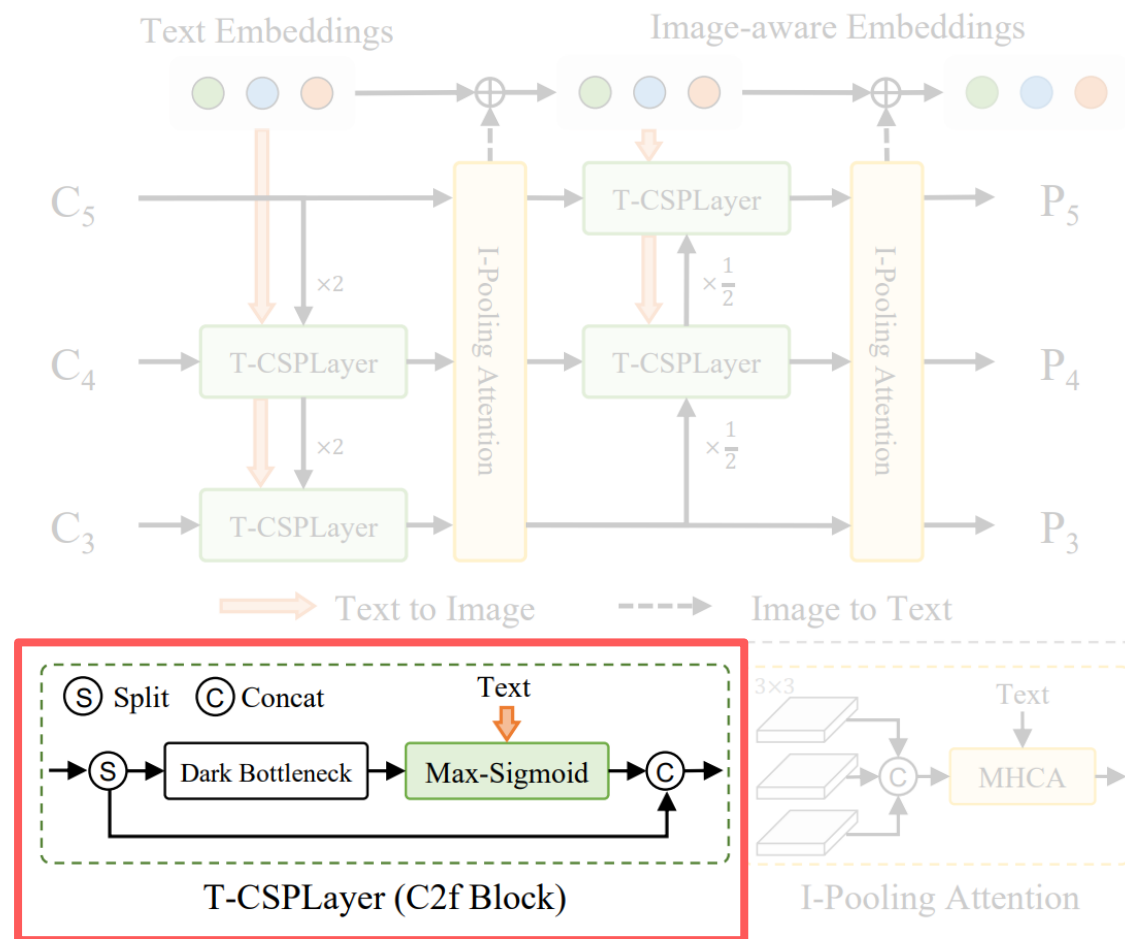
Text guided CSP layer

- 입력된 텍스트 정보를 이미지 특징에 통합
 - text-guidance를 multiple image features에서 진행

$$X'_l = X_l \cdot \delta\left(\max_{j \in \{1..C\}} (X_l W_j^T)\right)^T,$$

$$X_l \in \mathbb{R}^{H \times W \times D} \quad (l \in \{3, 4, 5\}),$$

$$W = \text{TextEncoder}(T) \in \mathbb{R}^{C \times D}$$

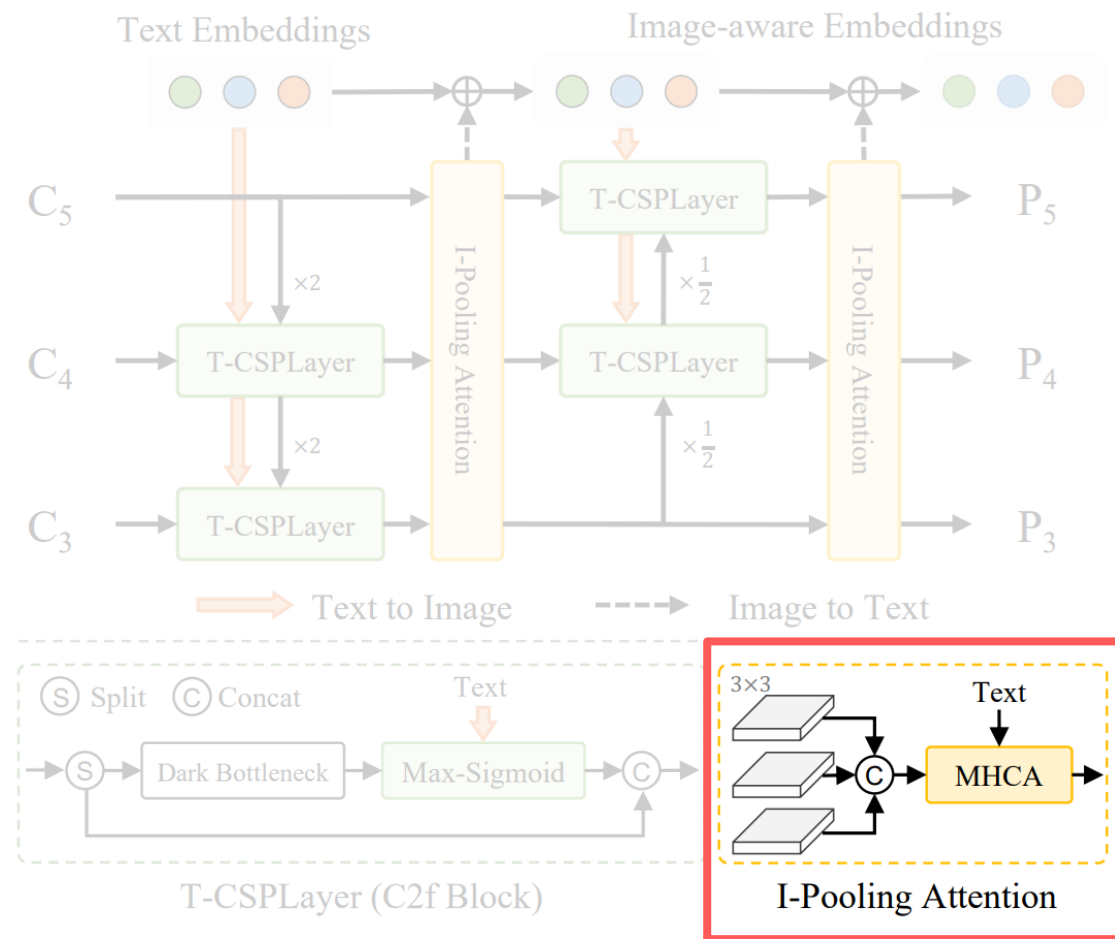


Method

Image Pooling Attention

- image features에 max pooling을 적용
- 3x3 region 3개를 얻음
 - $(3 \times 3) * 3(\text{multi-scale}) = 27$ 개의 token
- Text embedding 업데이트

$$W' = W + \text{MultiHead-Attention}(W, \tilde{X}, \tilde{X})$$



Method

- Pre-training schemes
 - Learning from Region-Text Contrastive Loss

$$\mathcal{L}(I) = \mathcal{L}_{\text{con}} + \lambda_I \cdot (\mathcal{L}_{\text{iou}} + \mathcal{L}_{\text{dfl}})$$

$L(I)$ 입력 이미지 I에 대한 총 Loss

L_{con} Region-text contrastive loss

L_{iou} IoU loss $\rightarrow$ Box regression loss

L_{dfl} Distributed focal loss $\rightarrow$ Box regression loss

λ_I Indicator factor, 이미지가 detection이나 grounding data에서 오면 1, image text data에서 오면 0

Method

- Pre-training schemes
 - Pseudo Labeling with Image-Text Data
 - Image-text pair를 pre-training에 바로 이용하는 대신 자동 labeling 진행
 - 1) 명사구 추출: n-gram 알고리즘 이용
 - 2) Pseudo labeling: 주어진 명사구에 대해 pre-trained OVD(e.g. GLIP)로 이미지 내 Pseudo boxes 생성
 - 3) Filtering: pre-trained CLIP을 사용, image-text pair, region-text pair의 관계를 구하여 낮은 것 제거
 - NMS 적용하여 중복되는 bbox 제거
 - CC3M 데이터에서 246k images 샘플링하여 821k pseudo labeling 진행

Experiments

Implementation

- MMYOLO 프레임워크 기반 (Baseline: YOLOv8)
- S/M/L 세 가지 모델 제공
- Open-source CLIP text encoder 사용 (input text → text embedding)
- 추론 속도는 NVIDIA V100 GPU 한대에서 측정
 - FP16, TensorRT와 같은 추가 가속 메커니즘 적용 X

Experiments

Pre-training

- Experimental Setup
 - AdamW
 - Initial learning rate: 0.002, weight decay: 0.05
 - 100 epoch, 32 NVIDIA V100 GPU
 - Batch size: 512
 - Data augmentation: color augmentation, random affine, random flip, mosaic with 4 images
 - CLIP text encoder는 frozen 상태로 pre-training

Experiments

Pre-training

- Pre-training Data
 - Objects365, GQA, Flickr, CC3M 등 대규모 데이터셋으로 pre-training

Dataset	Type	Vocab.	Images	Anno.
Objects365V1 [46]	Detection	365	609k	9,621k
GQA [17]	Grounding	-	621k	3,681k
Flickr [38]	Grounding	-	149k	641k
CC3M† [47]	Image-Text	-	246k	821k

Table 1. **Pre-training Data.** The specifications of the datasets used for pre-training YOLO-World.

- Zero-shot Evaluation: LVIS 데이터셋 (LVIS minival)
 - 최대 예측 개수는 1,000개로 설정

Experiments

AP (Average Precision): 전체 클래스에 대한 평균 정밀도 - 일반적인 detection 성능 지표
 AP_r (Rare): 희귀 클래스에 대한 AP
 AP_c (Common): 일반적인 클래스에 대한 AP
 AP_f (Frequent): 자주 나오는 클래스에 대한 AP

Method	Backbone	Params	Pre-trained Data	FPS	AP	AP _r	AP _c	AP _f
MDETR [21]	R-101 [15]	169M	GoldG	-	24.2	20.9	24.3	24.2
GLIP-T [24]	Swin-T [32]	232M	O365,GoldG	0.12	24.9	17.7	19.5	31.0
GLIP-T [24]	Swin-T [32]	232M	O365,GoldG,Cap4M	0.12	26.0	20.8	21.4	31.0
GLIPv2-T [59]	Swin-T [32]	232M	O365,GoldG	0.12	26.9	-	-	-
GLIPv2-T [59]	Swin-T [32]	232M	O365,GoldG,Cap4M	0.12	29.0	-	-	-
Grounding DINO-T [30]	Swin-T [32]	172M	O365,GoldG	1.5	25.6	14.4	19.6	32.2
Grounding DINO-T [30]	Swin-T [32]	172M	O365,GoldG,Cap4M	1.5	27.4	18.1	23.3	32.7
DetCLIP-T [56]	Swin-T [32]	155M	O365,GoldG	2.3	34.4	26.9	33.9	36.3
YOLO-World-S	YOLOv8-S	13M (77M)	O365,GoldG	74.1 (19.9)	26.2	19.1	23.6	29.8
YOLO-World-M	YOLOv8-M	29M (92M)	O365,GoldG	58.1 (18.5)	31.0	23.8	29.2	33.9
YOLO-World-L	YOLOv8-L	48M (110M)	O365,GoldG	52.0 (17.6)	35.0	27.1	32.8	38.3
YOLO-World-L	YOLOv8-L	48M (110M)	O365,GoldG,CC3M [†]	52.0 (17.6)	35.4	27.6	34.1	38.0

Table 2. **Zero-shot Evaluation on LVIS.** We evaluate YOLO-World on LVIS *minival* [21] in a zero-shot manner. We report the *Fixed AP* [4] for a fair comparison with recent methods. [†] denotes the pseudo-labeled CC3M in our setting, which contains 246k samples. The FPS is evaluated on one NVIDIA V100 GPU w/o TensorRT. The parameters and FPS of YOLO-World are evaluated for both the re-parameterized version (w/o bracket) and the original version (w/ bracket).

Experiments

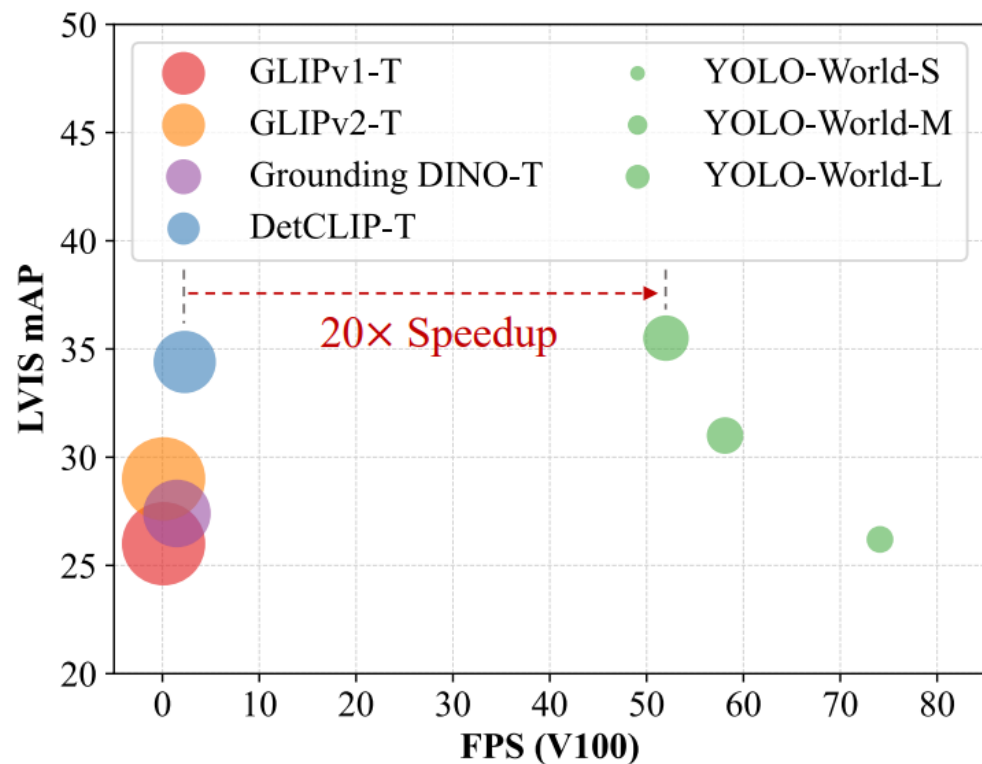


Figure 1. **Speed-and-Accuracy Curve.** We compare YOLO-World with recent open-vocabulary methods in terms of speed and accuracy. All models are evaluated on the LVIS_{minival} and inference speeds are measured on one NVIDIA V100 w/o TensorRT. The size of the circle represents the model’s size.

Experiments



Figure 5. **Visualization Results on Zero-shot Inference on LVIS.** We adopt the pre-trained YOLO-World-L and infer with the LVIS vocabulary (containing 1203 categories) on the COCO val2017.

Experiments

Ablation Experiments

- 다양한 pre-training 데이터

Pre-trained Data	AP	AP _r	AP _c	AP _f
O365	23.5	16.2	21.1	27.0
O365,GQA	31.9	22.5	29.9	35.4
O365,GoldG	32.5	22.3	30.6	36.0
O365,GoldG,CC3M [†]	33.0	23.6	32.0	35.5

Table 3. **Ablations on Pre-training Data.** We evaluate the zero-shot performance on LVIS of pre-training YOLO-World with different amounts of data.

Experiments

Ablation Experiments

- RepVL-PAN

GQA	T→I	I→T	AP	AP _r	AP _c	AP _f
✗	✗	✗	22.4	14.5	20.1	26.0
✗	✓	✗	23.2	15.2	20.6	27.0
✗	✓	✓	23.5	16.2	21.1	27.0
✓	✗	✗	29.7	21.0	27.1	33.6
✓	✓	✓	31.9	22.5	29.9	35.4

Table 4. **Ablations on Re-parameterizable Vision-Language Path Aggregation Network.** We evaluate the zero-shot performance on LVIS of the proposed Vision-Language Path Aggregation Network. T→I and I→T denote the Text-guided CSPLayers and Image-Pooling Attention, respectively.

Experiments

Ablation Experiments

- 텍스트 인코더 종류(BERT, CLIP 등)

Text Encoder	Frozen?	AP	AP _r	AP _c	AP _f
BERT-base	Frozen	14.6	3.4	10.7	20.0
BERT-base	Fine-tune	18.3	6.6	14.6	23.6
CLIP-base	Frozen	22.4	14.5	20.1	26.0
CLIP-base	Fine-tune	19.3	8.6	15.7	24.8

Table 5. **Text Encoder in YOLO-World.** We ablate different text encoders in YOLO-World through the zero-shot LVIS evaluation.

Experiments

Fine-tuning YOLO-World

- Experimental Setup
 - AdamW
 - Initial learning rate: 0.0002, lr of CLIP text encoder: 0.01
 - LVIS dataset의 경우 LVIS-base에서 fine-tuning 후 LVIS-novel에서 평가

Experiments

Fine-tuning YOLO-World

- COCO Object Detection

Method	Pre-train	AP	AP ₅₀	AP ₇₅	FPS
<i>Training from scratch.</i>					
YOLOv6-S [23]	✗	43.7	60.8	47.0	442
YOLOv6-M [23]	✗	48.4	65.7	52.7	277
YOLOv6-L [23]	✗	50.7	68.1	54.8	166
YOLOv7-T [52]	✗	37.5	55.8	40.2	404
YOLOv7-L [52]	✗	50.9	69.3	55.3	182
YOLOv7-X [52]	✗	52.6	70.6	57.3	131
YOLOv8-S [20]	✗	44.4	61.2	48.1	386
YOLOv8-M [20]	✗	50.5	67.3	55.0	238
YOLOv8-L [20]	✗	52.9	69.9	57.7	159
<i>Zero-shot transfer.</i>					
YOLO-World-S	O+G	37.6	52.3	40.7	-
YOLO-World-M	O+G	42.8	58.3	46.4	-
YOLO-World-L	O+G	44.4	59.8	48.3	-
YOLO-World-L	O+G+C	45.1	60.7	48.9	-
<i>Fine-tuned w/ RepVL-PAN.</i>					
YOLO-World-S	O+G	45.9	62.3	50.1	-
YOLO-World-M	O+G	51.2	68.1	55.9	-
YOLO-World-L	O+G+C	53.3	70.1	58.2	-
<i>Fine-tuned w/o RepVL-PAN.</i>					
YOLO-World-S	O+G	45.7	62.3	49.9	373
YOLO-World-M	O+G	50.7	67.2	55.1	231
YOLO-World-L	O+G+C	53.3	70.3	58.1	156

Table 6. **Comparison with YOLOs on COCO Object Detection.** We fine-tune the YOLO-World on COCO `train2017` and evaluate on COCO `val2017`. The results of YOLOv7 [52] and YOLOv8 [20] are obtained from MMYOLO [3]. ‘O’, ‘G’, and ‘C’ denote pertaining using Objects365, GoldG, and CC3M[†], respectively. The FPS is measured on one NVIDIA V100 w/ TensorRT.

Experiments

Fine-tuning YOLO-World

- LVIS Object Detection

Method	AP	AP _r	AP _c	AP _f
ViLD [13]	27.8	16.7	26.5	34.2
RegionCLIP [62]	28.2	17.1	-	-
Detic [63]	26.8	17.8	-	-
FVLM [22]	24.2	18.6	-	-
DetPro [8]	28.4	20.8	27.8	32.4
BARON [53]	29.5	23.2	29.3	32.5
YOLOv8-S	19.4	7.4	17.4	27.0
YOLOv8-M	23.1	8.4	21.3	31.5
YOLOv8-L	26.9	10.2	25.4	35.8
YOLO-World-S	23.9	12.8	20.4	32.7
YOLO-World-M	28.8	15.9	24.6	39.0
YOLO-World-L	34.1	20.4	31.1	43.5

Table 7. **Comparison with Open-Vocabulary Detectors on LVIS.** We train YOLO-World on the LVIS-base (including common and frequent) report the *bbox AP*. The YOLO-v8 are trained on the full LVIS datasets (including base and novel) along with the class balanced sampling.

Experiments

Open-vocabulary Instance Segmentation

- COCO to LVIS: COCO로 fine-tuning, LVIS로 평가 (80 $\rightarrow$ 1203)
- LVIS to COCO: LVIS-base로 fine-tuning, LVIS로 평가 (866 $\rightarrow$ 1203)

Model	Fine-tune Data	Fine-tune Modules	AP	AP _r	AP _c	AP _f	AP ^b	AP _r ^b
YOLO-World-M	COCO	<i>Seg Head</i>	12.3	9.1	10.9	14.6	22.3	16.2
YOLO-World-L	COCO	<i>Seg Head</i>	16.2	12.4	15.0	19.2	25.3	18.0
YOLO-World-M	LVIS-base	<i>Seg Head</i>	16.7	12.6	14.6	20.8	22.3	16.2
YOLO-World-L	LVIS-base	<i>Seg Head</i>	19.1	14.2	17.2	23.5	25.3	18.0
YOLO-World-M	LVIS-base	<i>All</i>	25.9	13.4	24.9	32.6	32.6	15.8
YOLO-World-L	LVIS-base	<i>All</i>	28.7	15.0	28.3	35.2	36.2	17.4

Table 8. **Open-Vocabulary Instance Segmentation.** We evaluate YOLO-World for open-vocabulary instance segmentation under the two settings. We fine-tune the segmentation head or all modules of YOLO-World and report *Mask AP* for comparison. AP^b denotes the box AP.

Experiments



{men, women, boy, girl} {elephant, ear, leg, trunk, ivory} {golden dog, black dog, spotted dog} {grass, sky, zebra, trunk, tree}

Figure 6. **Visualization Results on User’s Vocabulary.** We define the custom vocabulary for each input image and YOLO-World can detect the accurate regions according to the vocabulary. Images are obtained from COCO val2017.

Experiments



Figure 7. **Visualization Results on Referring Object Detection.** We explore the capability of the pre-trained YOLO-World to detect objects with descriptive noun phrases. Images are obtained from COCO val2017.

Conclusion

- YOLO를 vision-language model로 재설계
- 속도와 open-vocabulary 성능 동시 달성
- 소규모 모델에서 vision-language 사전 훈련 효과 최초 입증
- 실제 환경에서 즉시 적용 가능

Q&A

